CHAPTER 2

The Building Blocks: Curves, Discount Factors, Forward Rates

2.1 Money now versus money later

Every instrument in this book, however exotic, reduces to a portfolio of promises to pay money at specified future dates. To value such a promise you need exactly one primitive object: the price today of one unit of currency delivered at a future date T, with certainty. This is the **discount factor**, written $P(t, T)$ — the price at time t of a **zero-coupon bond** maturing at T, paying 1 with no default risk.

> **Key result: the discount factor**
>
> $P(t, T)$ is the value at time t of \$1 received for certain at time $T \geq t$. By construction $P(T, T) = 1$, and in a normal (non-negative-rate, positively-priced-money) market $0 < P(t, T) \leq 1$ for $T > t$: money later is worth no more than money now. $P(t, \cdot)$, viewed as a function of T, is the **discount curve**.

A discount factor is not directly quoted in the market. What trades are instruments — deposits, futures, swaps — whose prices imply a discount curve. **Bootstrapping** is the procedure of inverting a ladder of liquid instrument prices, from shortest maturity to longest, into a discount factor at each corresponding maturity, then interpolating between them. This book will not dwell on the mechanics of curve-building, which is a substantial topic of its own, but every later chapter takes it as given that a discount curve $P(0, T)$ exists and is available at any T needed.

2.2 From discount factors to rates

A discount factor is inconvenient to think about directly — nobody quotes "$P(0, 5) = 0.8320$" at a trading desk. Rates are the human-readable transform of the same information.

Simple (money-market) compounding

For a period $[T_1, T_2]$, the **simple forward rate** $L(T_1, T_2)$ is defined so that investing $P(t, T_1)$ at t, rolled forward risk-free to T_1 and then invested at the fixed rate L over $[T_1, T_2]$, exactly reproduces $P(t, T_2)$:

> **Key result: the simple forward rate**
>
> $$1+\tau(T_1, T_2)\, L(t; T_1, T_2) = \frac{P(t, T_1)}{P(t, T_2)} \qquad \Longrightarrow \qquad L(t; T_1, T_2) = \frac{1}{\tau(T_1, T_2)} \left(\frac{P(t, T_1)}{P(t, T_2)} - 1 \right)$$
>
> where $\tau(T_1, T_2)$ is the year fraction of the period under the relevant day-count convention (e.g. Actual/360 for USD money markets, Actual/365 for GBP). This is the object that both LIBOR and its replacement rates approximate, and it is the fundamental state variable of the entire BGM model built in Part IV.

Setting $t = T_1$ recovers the familiar spot rate: $L(T_1; T_1, T_2)$ is the rate observed *at* T_1 that applies to the period ending at T_2. For $t < T_1$, $L(t; T_1, T_2)$ is a *forward-looking* rate: the market's current expectation, embedded in today's curve, of what the rate over a future period will turn out to be. This distinction — a rate fixed in advance of its accrual period versus fixed only after it — turns out to matter enormously once SOFR replaces LIBOR, and Chapter 3 is built entirely around it.

Worked example 2.1 — extracting a forward rate from discount factors.
A SOFR discount curve gives $P(0, 5) = 0.8285$ and $P(0, 5.25) = 0.8210$ (five years and five-and-a-quarter years respectively). Find the 3-month forward rate for the period $[5, 5.25]$, using Actual/360 with $\tau = 0.25$ exactly for simplicity.
Step 1 — apply the key result.

$$L(0; 5, 5.25) = \frac{1}{0.25}\left(\frac{0.8285}{0.8210} - 1\right) = \frac{1}{0.25}(1.009135 - 1) = \frac{0.009135}{0.25} = 0.03654$$

Step 2 — express as a rate.

$$L(0; 5, 5.25) = 3.654\%$$

Sanity check. The ratio $P(0, 5)/P(0, 5.25) = 1.009135$ is only slightly above 1, as expected for a short 3-month period at a mid-single-digit percentage rate: over 0.25 years at roughly 3.65%, compounding should add close to $0.25 \times 3.65\% \approx 0.91\%$, which matches.
Interpretation. This is the forward-looking 3-month SOFR rate the market expects to prevail five years from now, as implied by today's curve — not a forecast in the sense of a point prediction, but the rate at which a bank could lock in that period's borrowing cost today using instruments that trade now. This exact rate, $F = 3.654\%$ maturing in five years, is the object carried forward as a running example through Chapters 5, 6 and 13.

2.3 Continuous compounding and the short rate

A second convention, used more in theoretical derivations than in trading conversation, is **continuous compounding**: $P(t, T) = e^{-r(t,T)(T-t)}$ for a continuously compounded rate $r(t, T)$, or in the limit of infinitesimal maturity, the **instantaneous forward rate**

$$f(t, T) = -\frac{\partial}{\partial T} \log P(t, T).$$

This object is the state variable of the Heath–Jarrow–Morton framework built in Chapter 10, and it is worth flagging now, before it is needed, that $f(t, T)$ is a theoretical limit — an infinitesimally short forward rate — and is not itself a traded quantity. The tension between working with this elegant but unobservable continuous object (HJM) and working directly with the clunkier but market-observable simple forward rates of this section (BGM) is, in a real sense, the plot of Part III.

PITFALL

Continuously compounded rates and simply compounded (money-market) rates are numerically close but not identical, and conflating them silently produces small, hard-to-diagnose pricing errors that compound (no pun intended) across a long-dated instrument. If a discount factor is computed via e^{-rT} when the rate r was quoted on a simple Actual/360 basis, the error is typically tens of basis points in the resulting discount factor for a 10-year maturity — small enough to pass a casual sanity check, large enough to fail a P&L reconciliation. Always carry the day-count convention alongside the rate, not just the number.

2.4 The curve as a function, and why it moves

A discount curve is not a static object frozen at $t = 0$; it is a snapshot at each instant t of the market's current view of the cost of money across every future horizon. As new information arrives — a central bank decision, an inflation print, a shift in risk appetite — every point on the curve can move, and the *whole curve* generally moves together but not by the same amount at every maturity: short rates react more sharply to policy surprises, long rates more to growth and inflation expectations. This is precisely why a rates option is genuinely harder to model than a single-stock option: the underlying is not one number but an entire curve, and a full model of the rates market must describe how every point on that curve moves, together, consistently, without arbitrage. That is exactly what BGM does, and it is the reason the model needs a whole book's worth of machinery that a single-asset option model does not.

ON THE DESK

Curve-building is treated by most trading desks as solved, boring infrastructure — until it silently breaks, at which point it becomes the most urgent problem in the building. A stale instrument in the bootstrap, a wrong day-count convention on one leg, or an interpolation method that produces a curve with visible kinks (and therefore, oscillating forward rates) can corrupt every price computed downstream, including every caplet, swaption and calibration output built on top of it. A recalibration engine should always validate its input curve — checking that forward rates implied by the curve are smooth and economically plausible — before spending any computation trying to calibrate a volatility model to prices built on a broken one. Chapter 20 returns to this as the first line of defence in a production system.

Chapter benchmark

Before moving on, you should be able to, without notes:

1. State the definition of a discount factor $P(t, T)$ and explain why it typically lies between 0 and 1.
2. Derive the simple forward rate $L(t; T_1, T_2)$ from two discount factors and a day-count fraction, and compute it numerically.
3. Explain the difference between a forward-looking rate (fixed in advance) and a rate that can only be known after its accrual period ends, and why this distinction will matter once RFRs are introduced.